

EXPERIMENT 10

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Flowchart, Pseudocode, Algorithm, Test Cases, C Code, Compilation and Execution

Questions: (1) Convert an input time in seconds into days, hours, minutes and seconds. (2) Input a year and determine whether it is a leap year.

1. Convert Seconds into Days, Hours, Minutes and Seconds

Objective: Read a non-negative time value in seconds and convert it into days, hours, minutes, and remaining seconds.

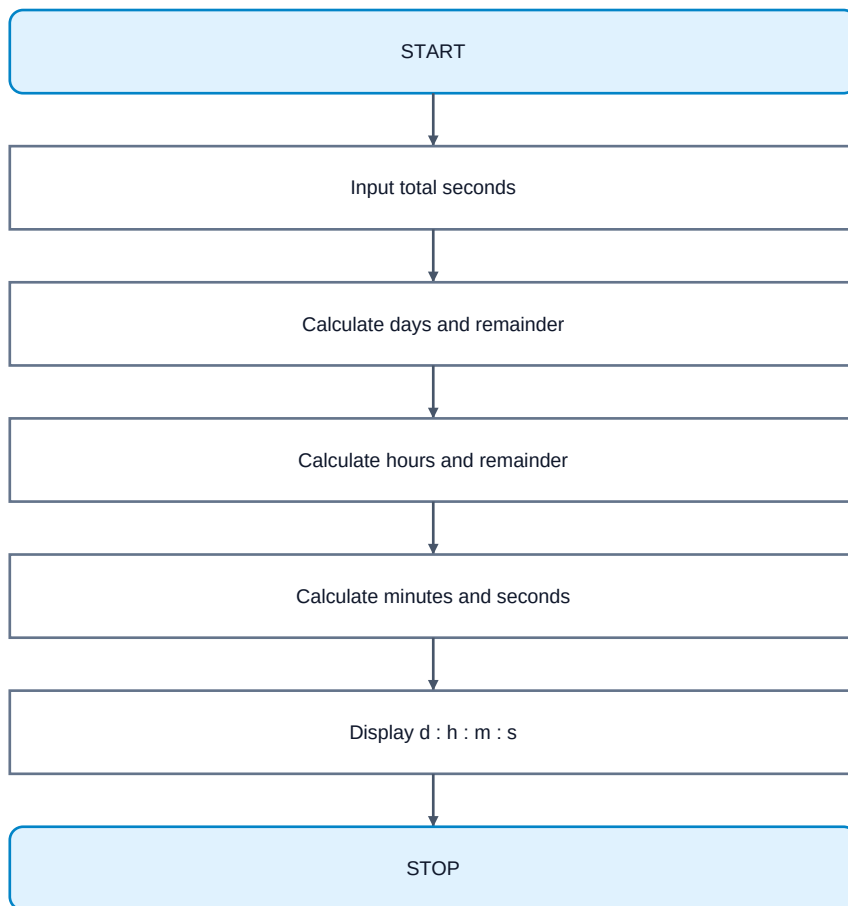
Algorithm

1. Start.
2. Input total seconds.
3. $\text{days} = \text{total} / 86400$.
4. $\text{remaining} = \text{total} \% 86400$.
5. $\text{hours} = \text{remaining} / 3600$; $\text{remaining} = \text{remaining} \% 3600$.
6. $\text{minutes} = \text{remaining} / 60$.
7. $\text{seconds} = \text{remaining} \% 60$.
8. Display days, hours, minutes and seconds.
9. Stop.

Pseudocode

```
START
INPUT totalSeconds
days <- totalSeconds / 86400
remaining <- totalSeconds % 86400
hours <- remaining / 3600
remaining <- remaining % 3600
minutes <- remaining / 60
seconds <- remaining % 60
PRINT days, hours, minutes, seconds
STOP
```

Flowchart



Test Case Table

Test	Input seconds	Expected output
1	60	0 days, 0 hours, 1 minute, 0 seconds
2	3661	0 days, 1 hour, 1 minute, 1 second
3	90061	1 day, 1 hour, 1 minute, 1 second
4	172800	2 days, 0 hours, 0 minutes, 0 seconds
5	0	0 days, 0 hours, 0 minutes, 0 seconds

C Program

```
#include <stdio.h>

int main() {
    long long total, days, hours, minutes, seconds, remaining;

    printf("Enter time in seconds: ");
    scanf("%lld", &total);

    if (total < 0) {
        printf("Time cannot be negative.\n");
        return 0;
    }

    days = total / 86400;
    remaining = total % 86400;
    hours = remaining / 3600;
    remaining = remaining % 3600;
    minutes = remaining / 60;
    seconds = remaining % 60;
```

```
    printf("%lld days, %lld hours, %lld minutes, %lld seconds\n",  
           days, hours, minutes, seconds);  
    return 0;  
}
```

Compile and Execute

```
gcc time_convert.c -o time_convert  
./time_convert
```

Sample Execution

```
Enter time in seconds: 90061  
1 days, 1 hours, 1 minutes, 1 seconds
```

2. Check Whether a Year is a Leap Year

Objective: Input a year and determine whether it is a leap year according to the Gregorian calendar rules.

Leap Year Rule

A year is a leap year if it is divisible by 400, or if it is divisible by 4 but **not** divisible by 100. Thus 2000 is a leap year, while 1900 is not.

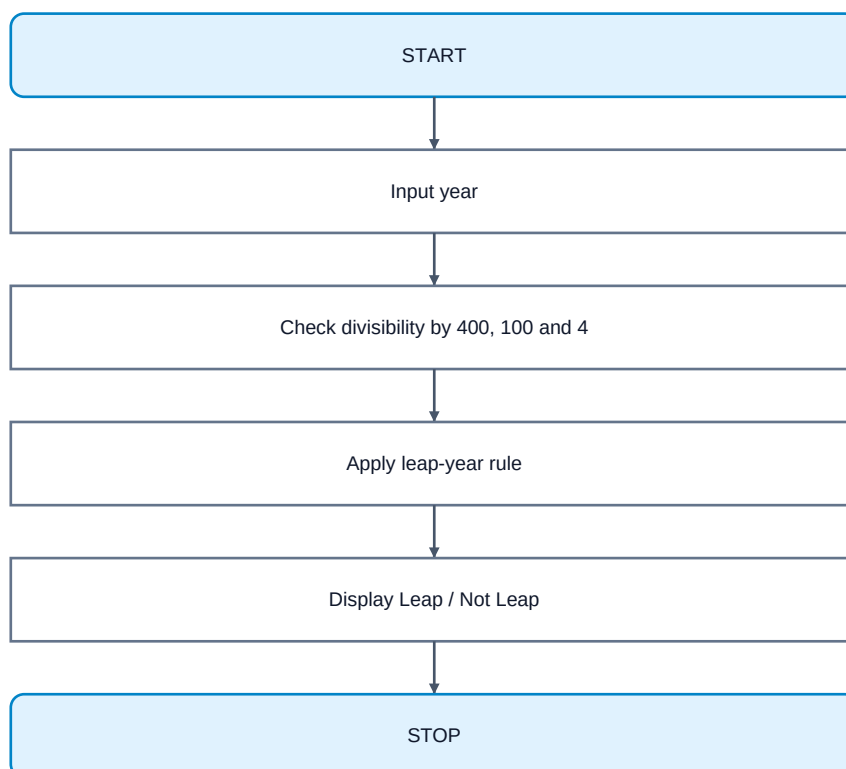
Algorithm

1. Start.
2. Input year.
3. If year is divisible by 400, mark it as leap year.
4. Otherwise, if year is divisible by 100, mark it as not a leap year.
5. Otherwise, if year is divisible by 4, mark it as leap year.
6. Otherwise, mark it as not a leap year.
7. Display the result.
8. Stop.

Pseudocode

```
START
INPUT year
IF year % 400 = 0 THEN
    PRINT "Leap year"
ELSE IF year % 100 = 0 THEN
    PRINT "Not a leap year"
ELSE IF year % 4 = 0 THEN
    PRINT "Leap year"
ELSE
    PRINT "Not a leap year"
END IF
STOP
```

Flowchart



Test Case Table

Test	Year	Expected result	Reason
1	2024	Leap year	Divisible by 4, not by 100
2	2023	Not leap year	Not divisible by 4
3	2000	Leap year	Divisible by 400
4	1900	Not leap year	Divisible by 100, not by 400
5	2400	Leap year	Divisible by 400

C Program

```
#include <stdio.h>

int main() {
    int year;

    printf("Enter a year: ");
    scanf("%d", &year);

    if (year % 400 == 0)
        printf("%d is a leap year.\n", year);
    else if (year % 100 == 0)
        printf("%d is not a leap year.\n", year);
    else if (year % 4 == 0)
        printf("%d is a leap year.\n", year);
    else
        printf("%d is not a leap year.\n", year);

    return 0;
}
```

Compile and Execute

```
gcc leap_year.c -o leap_year
./leap_year
```

Sample Execution

```
Enter a year: 2024
2024 is a leap year.
```

Testing Summary

The prepared test cases cover ordinary values and important boundary rules. For time conversion, zero, minute/hour/day boundaries, and a mixed value are tested. For leap years, normal years, years divisible by 4, century years, and years divisible by 400 are tested.